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Graduate School

CARTESIAN *Worldview*

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Presenter

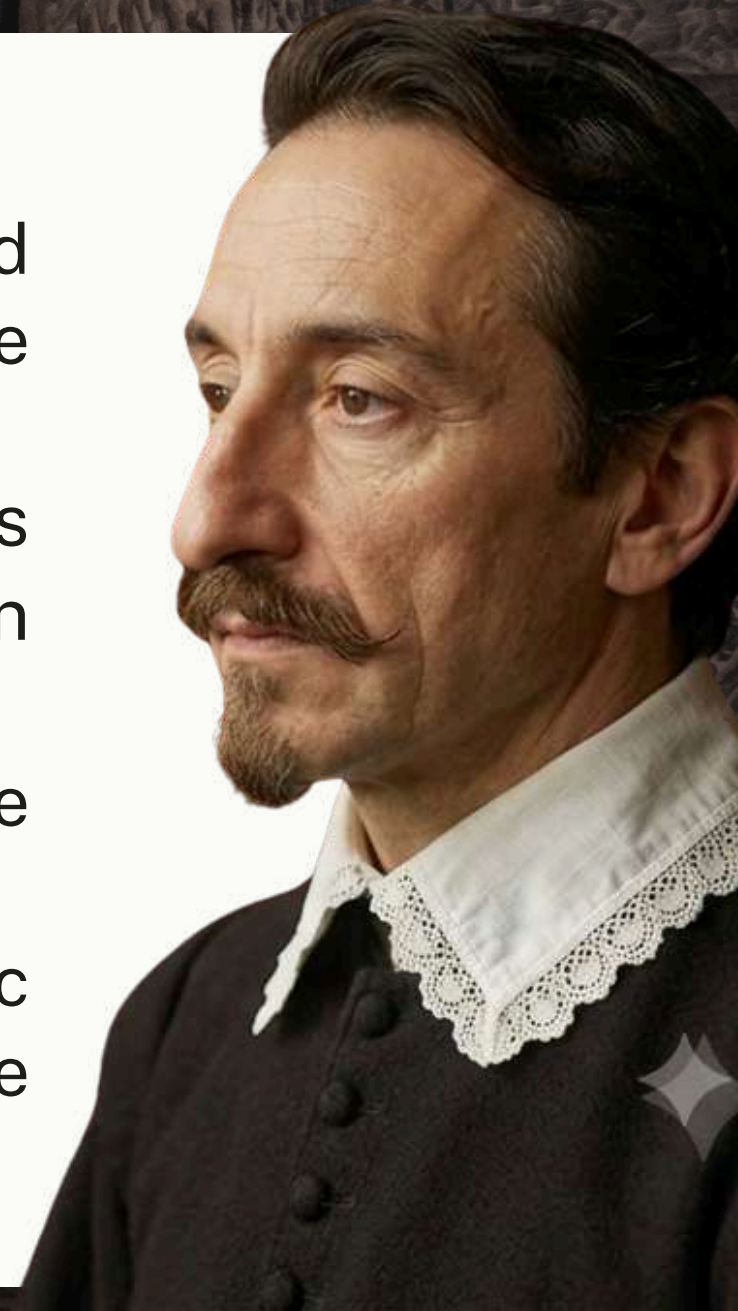




Learning Objectives

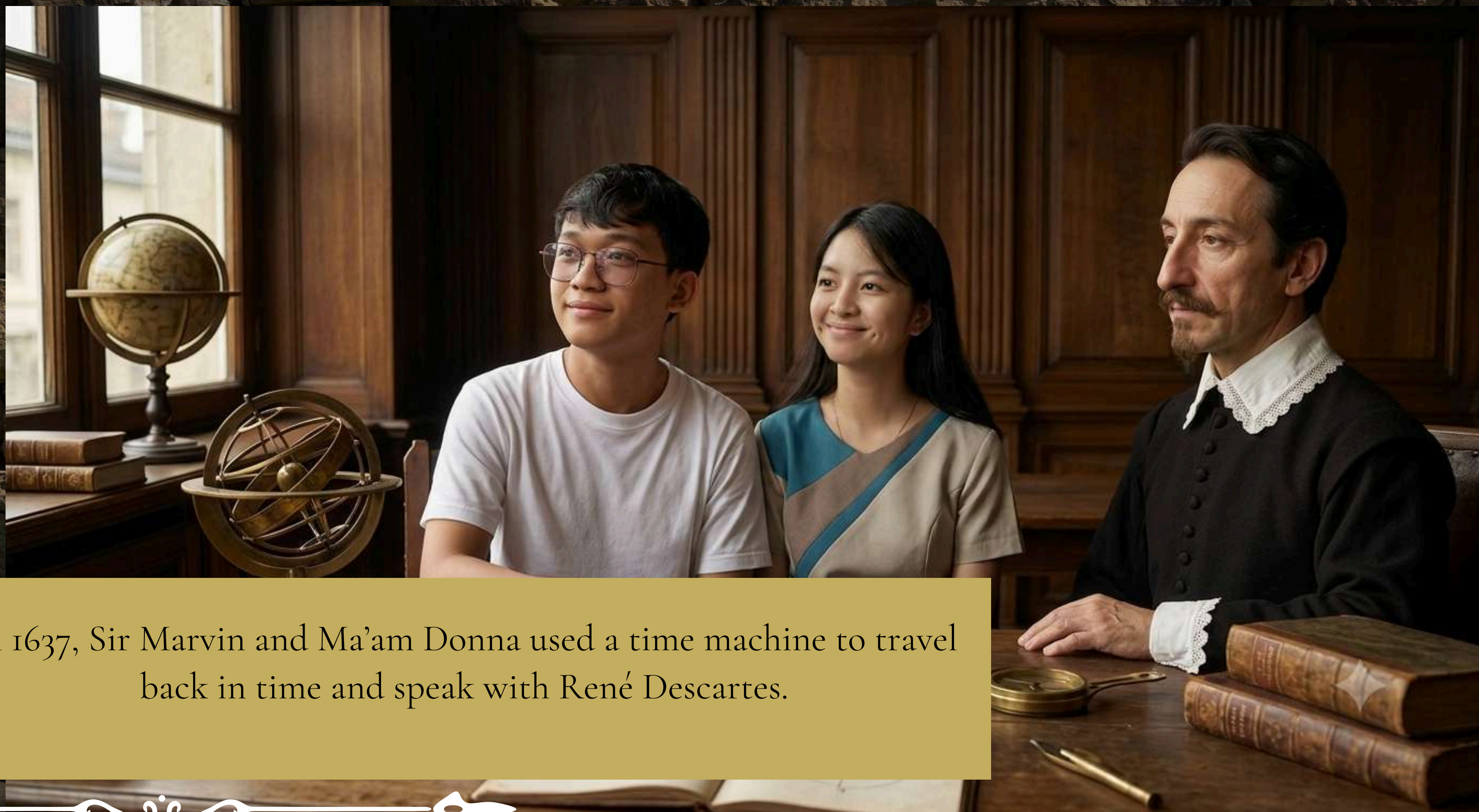
At the end of the lesson, the learners should:

1. Define mind as a thinking substance and matter as an extended substance, and deconstruct how Descartes uses radical doubt, cogito ergo sum, and the ontological argument to establish this dualistic reality.
2. Explain how treating the body as a machine driven by animal spirits explains biological functions while creating the unresolved mind-body interaction dilemma.
3. Apply Descartes' first two laws of motion to physical phenomena to demonstrate how the principle of action by contact completely replaces Aristotelian teleology.
4. Analyze how the transition to a structurally complex worldview shifted scientific evaluation away from Aristotelian intuition toward the hypothetico-deductive method and the universal application of mathematics.





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In 1637, Sir Marvin and Ma'am Donna used a time machine to travel back in time and speak with René Descartes.





1. Historical Context and the Crisis of the 17th Century

The seventeenth century represents a profound epistemic breach in the history of Western thought, a period of transition where the Aristotelian-Medieval worldview faced an existential crisis.



Galileo Galilei

Heliocentrism



Robert Boyle

Boyle's Law



Pierre Gassendi

Epicurean
atomism



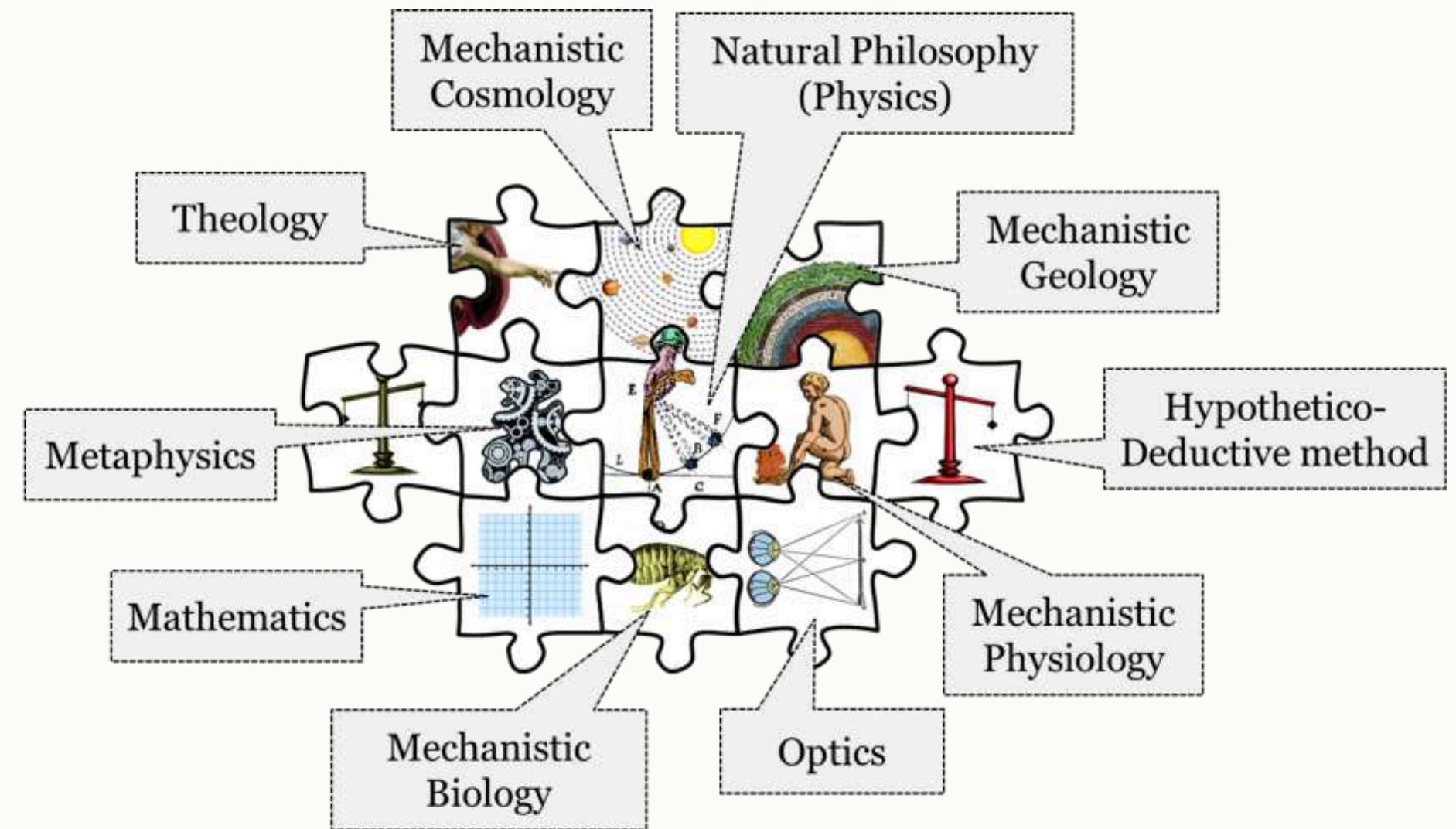
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Are there scientific ideas today that face similar resistance because they don't fit into current frameworks?



1. Historical Context and the Crisis of the 17th Century

René Descartes' strategic triumph lay in his architectonic approach: he did not merely propose new theories but constructed a complete "mosaic" that replaced the medieval system in toto.

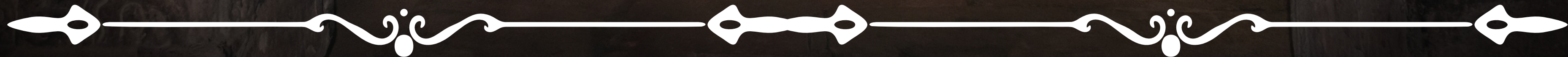




1. Historical Context and the Crisis of the 17th Century

This shift resulted in a distinct geopolitical intellectual divide. While the "Isles" (Britain) eventually moved toward a Newtonian synthesis, the "Continent"—including France, the Netherlands, and Sweden—adopted the Cartesian mosaic between 1700 and 1740.

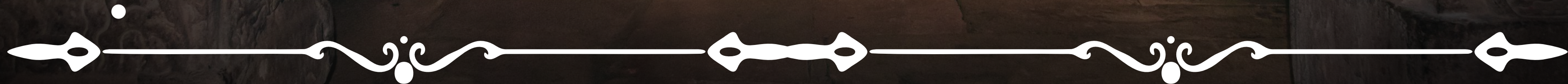
Feature	Aristotelian-Medieval Worldview	Cartesian Worldview (The Continent)
Dominance	Prevalent until the late 1600s	Accepted approx. 1700–1740
Core Structure	Pluralistic and Teleological	Mechanistic and Dualistic
Primary Method	Aristotelian-Medieval Method	Hypothetico-Deductive Method
Ontology	Hylomorphic (Matter + Form)	Mechanomorphic (Matter as Extension)





The Strategic Success of the Systematic Mosaic

- Descartes' success was rooted in the comprehensiveness of his project. By offering a totalizing system, encompassing metaphysics, physics, and physiology, he filled the historical vacuum left by the decaying medieval consensus. Where others offered fragments, Descartes offered a replacement world. This systematic replacement provided the necessary stability for scientific communities to abandon the medieval status quo, creating a new foundation for the inquiry into nature.





2. Epistemological Foundations: Radical Doubt and the Cogito

- Descartes recognized that any enduring scientific edifice required an unshakable starting point. To move beyond the crumbling authority of the past, he initiated a process of radical doubt, designed not as an end-point of skepticism, but as a methodological tool to identify a "bedrock" of certainty upon which a new intellectual history could be written.

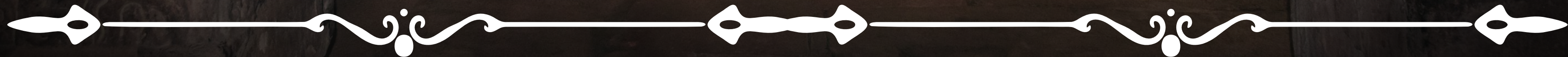


2. Epistemological Foundations: Radical Doubt and the Cogito

The Three-Stage Process of Radical Doubt

1. The Unreliable Witness (Sensory Doubt)

- Descartes argued that because our senses frequently deceive us, no observation-based knowledge can be considered infallible.



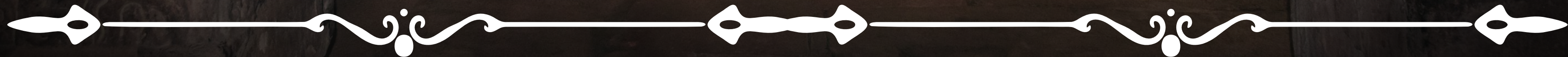


2. Epistemological Foundations: Radical Doubt and the Cogito

The Three-Stage Process of Radical Doubt

2. The Dream Hypothesis

- Descartes pushes his skepticism further by questioning how we can definitively distinguish a waking state from a dream.





2. Epistemological Foundations: Radical Doubt and the Cogito

The Three-Stage Process of Radical Doubt

3. The Evil Demon Hypothesis (Deceiving God)

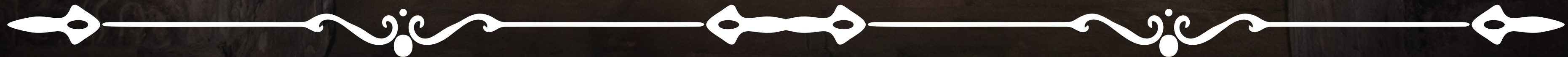
- He contended that since even the most learned experts are prone to error, all traditional education and external testimony must be set aside.



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*"I think, therefore I
am"*





2. Epistemological Foundations: Radical Doubt and the Cogito

The Logical Deduction: Cogito, Ergo Sum

Through this absolute negation, Descartes arrived at a singular, indubitable truth: the act of doubt itself necessitates a subject.

- ***I Doubt:*** Even under the influence of a deceiver, the act of doubting is occurring.
- ***Therefore, I Think:*** Doubting is an irreducible mode of cognition.
- ***Therefore, I Am:*** Cogito ergo sum—thinking necessitates the existence of a thinking mind.



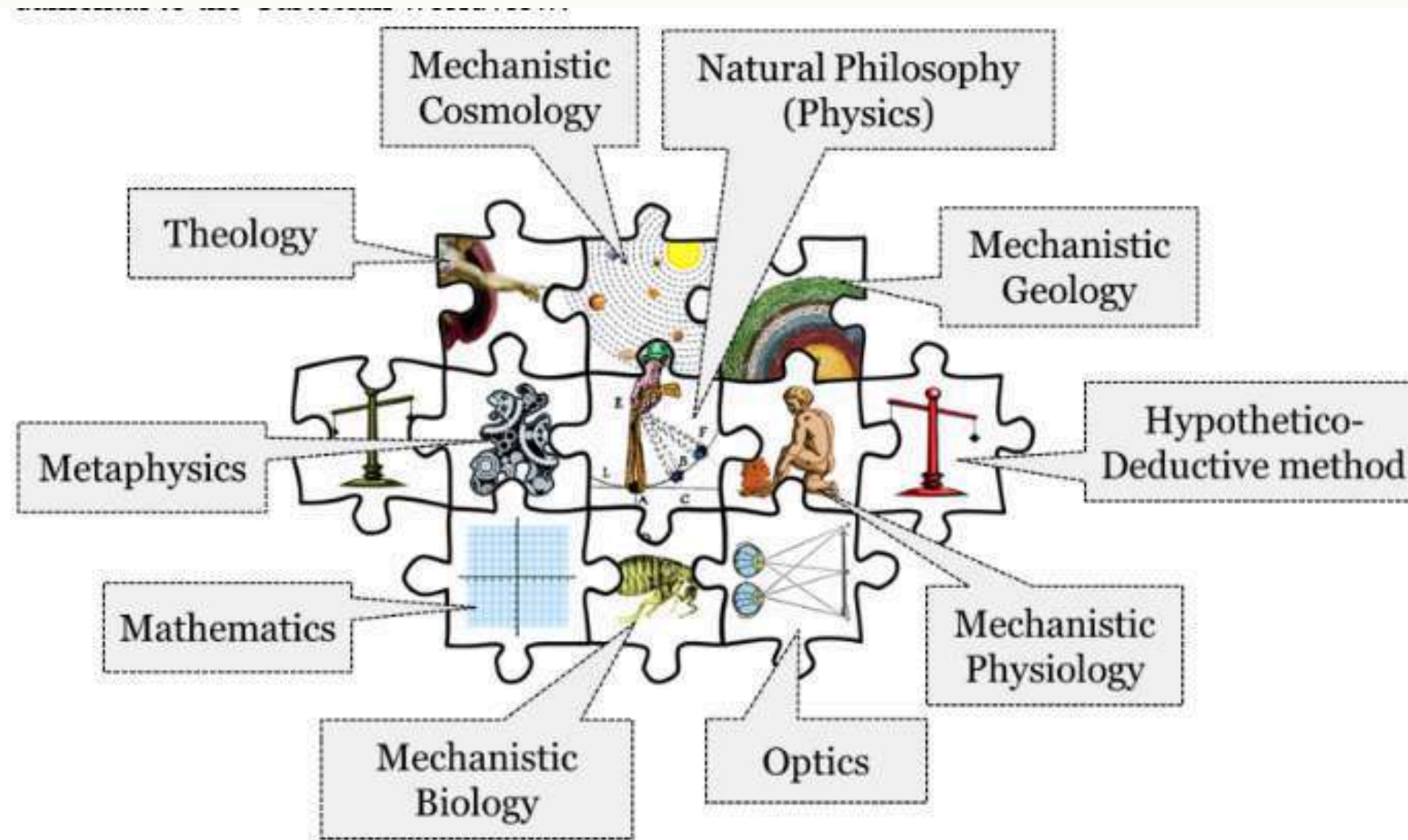
2. Epistemological Foundations: Radical Doubt and the Cogito

The Shift to the Subjective Foundation

- This deduction marked a revolutionary shift in the history of science: the foundation of knowledge was moved from external, traditional authority to the interiority of the individual thinking mind. However, this established only the existence of the incorporeal mind; the existence of a corporeal, external world still required a logical bridge.



3. The Theological Guarantor and the Ontological Argument



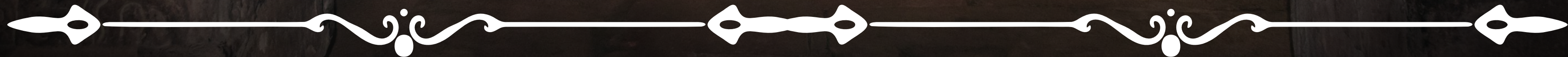
In the ***Cartesian Mosaic***, theology was not a peripheral concern but the central "***guarantor***" of objective reality. Descartes' physics and metaphysics were inextricably linked to his conception of the divine, as only a specific type of deity could bridge the gap between internal thought and external matter.



3. The Theological Guarantor and the Ontological Argument

The Involuntariness Argument and the External World

- Before proving the nature of the world, Descartes first sought to prove its existence through the **Involuntariness Argument**. He observed that certain sensations, most notably the feeling of heat from a fire, enter the mind regardless of his will. Because these sensations are independent of the mind's control, they must originate from a source external to the mind. This established that "something" exists outside the self.





3. The Theological Guarantor and the Ontological Argument

The Ontological Argument and the Veracious God

To prove that this external source is material and not a deception, Descartes employed the Ontological Argument:

1. God is defined as a "supremely perfect being," possessing all perfections: omnipotence, benevolence, and, crucially veracity (truth-telling).
2. Descartes argued that he clearly and distinctly perceived that non-existence is an imperfection.
3. Therefore, a supremely perfect being must necessarily exist.



3. The Theological Guarantor and the Ontological Argument

The Philosophical Bridge to Extension

- The significance of this theological pivot lies in its role as a guarantor of science. Because God is veracious and benevolent, He would not grant us a clear and distinct belief in a material world if that world were an illusion. Thus, the external world is confirmed to be a realm of extension — substance that occupies space.



4. The Metaphysical Pivot: From Hylomorphism to Mechanicism

1. Mechanicism vs. Hylomorphism

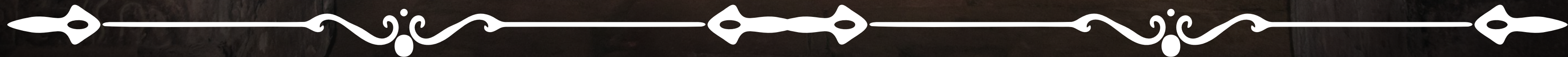
- **Cartesian View:** Rejection of "substantial forms" in favor of the view that all material things are merely interacting parts defined by extension.
- **Aristotelian View:** Belief that every substance is a composite of matter and an organizing principle or "form" (soul).



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If Descartes believed in God as a supremely perfect and truth-telling being, why did he still focus on mechanicism, the view that matter is only extension and machine-like, rather than the Aristotelian idea of matter + soul?"

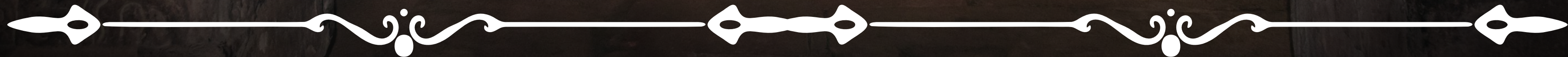




4. The Metaphysical Pivot: From Hylomorphism to Mechanicism

2. Dualism vs. Pluralism

- **Cartesian View:** Reduction of the universe to only two substances: Res Cogitans (Mind/Thought) and Res Extensa (Matter/Extension).
- **Aristotelian View:** A plurality of substances as numerous as the types of things in nature (e.g., "lion-ness" or "stone-ness").

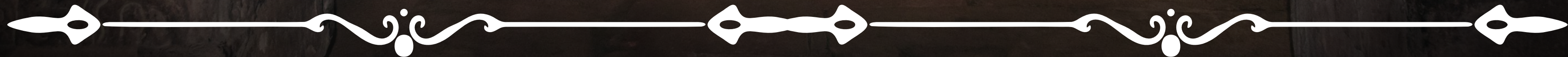




4. The Metaphysical Pivot: From Hylomorphism to Mechanicism

3. Action by Contact vs. Teleology

- **Cartesian View:** All change results from mechanical collision — "action by contact."
- **Aristotelian View:** Movement is "goal-oriented," driven by a "final cause" or purpose intrinsic to the object.





4. The Metaphysical Pivot: From Hylomorphism to Mechanicism

The Mathematization of Nature

- The "**So What?**" of this shift is the total mechanomorphism of the universe. By removing "**purpose**" and "**soul**" from matter and reducing it to **geometry and collision**, Descartes mathematized nature. The universe ceased to be a living organism striving toward goals and became a vast machine, governed by the laws of motion rather than the whims of forms.



5. Cartesian Physiology: The Body as a Machine

- Applying his mechanistic metaphysics to the living world, Descartes' *Treatise on Man* rejected the Aristotelian "humoral theory" in favor of a strictly mechanical anatomy. If the body is an extended substance, it must function via action by contact.



5. Cartesian Physiology: The Body as a Machine

Descartes described physiological processes as mechanical interactions:

- The Senses: Smell and sight were explained as particles of matter colliding with the senses, mechanically transmitting "tugs" along internal threads to the brain.**



5. Cartesian Physiology: The Body as a Machine

Descartes described physiological processes as mechanical interactions:

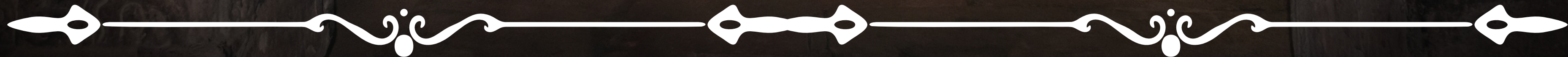
- The Heart as Furnace: Rejection of the heart as a mystical seat of life; it was redefined as a heat-source that caused blood to expand and circulate through purely thermal-mechanical means.**



5. Cartesian Physiology: The Body as a Machine

The Role of Material Fluidity

- **Animal Spirits:** Descartes identified a "fiery, airy fluid" known as animal spirits. Crucially, these were defined as a material fluid, not a spiritual one.
- **Mechanical Motion:** Muscles contracted when inflated by these material spirits, directed by a complex system of mechanical valves within the nerves.

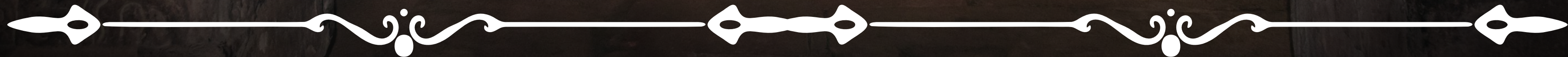




5. Cartesian Physiology: The Body as a Machine

The Philosophical Implications of Corporeal Mechanicism

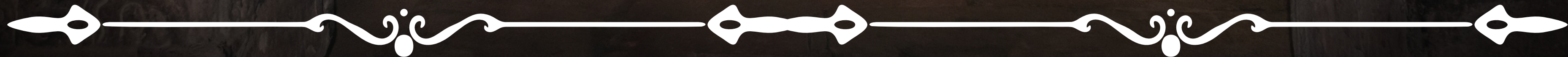
- **The "body-as-machine" concept allowed for biological processes like digestion, reflexes, and even basic emotions to be explained without recourse to a "soul." This stripped the corporeal world of its mystery, allowing biology to be studied with the same rigor as mechanics.**





6.. The Mind-Body Problem: Citizens of Two Worlds

- The inherent conflict of Cartesian dualism lies in the coordination of two substances, mind and matter, that share no common properties. This created a lasting mystery regarding how an immaterial command can move a material arm.





6.. The Mind-Body Problem: Citizens of Two Worlds

Proposed Solution	Mechanism	Philosophical Logic
Interactionism	The Pineal Gland	A causal influence where "will" converts into particle motion within a specific gland in the brain.
Parallelism	Pre-established Harmony	Mind and matter do not interact but are synchronized like two clocks by a divine clockmaker (God).



6.. The Mind-Body Problem: Citizens of Two Worlds

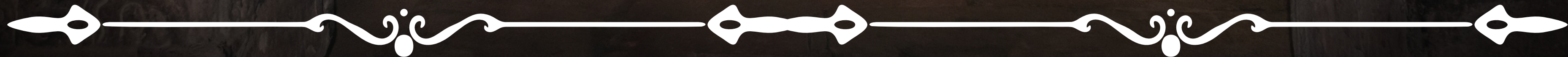
The Resilience of the Cartesian Synthesis

- The failure to solve the "**Mind-Body Problem**" is a critical historical insight. While the pineal gland explanation was widely viewed as insufficient, the Cartesian mosaic remained dominant. The community tolerated this unresolved mystery because the mechanistic explanations for the rest of the physical and physiological world were too powerful and logically consistent to abandon.



1. Conclusion: The Significance of the Cartesian Mosaic

- The Cartesian worldview represent the first successful, comprehensive challenge to the medieval status quo. Though its dominance on the European continent was relatively brief (1700–1740), it fundamentally altered the trajectory of human thought.

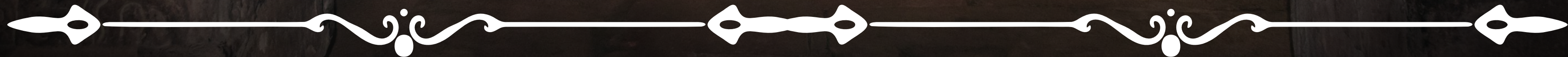




1. Conclusion: The Significance of the Cartesian Mosaic

Descartes' legacy is defined by **four core principles** that dismantled Aristotelianism:

1. Thought as the defining attribute of the mind.
2. Extension as the defining attribute of matter.
3. Mechanicism as the explanatory framework for the material world.
4. Dualism as the separation of the thinking self from the mechanical environment.





1. Conclusion: The Significance of the Cartesian Mosaic

- Ultimately, Descartes' work paved the way for the future of physics and biology by transforming the universe into a machine. He left us with the enduring image of the human being as a **"citizen of two worlds"—an immaterial, thinking mind piloting a mechanical, extended body.** This tension remains one of the most significant and lasting legacies in intellectual history.



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Cartesian Physics *and Cosmology*





**WHAT HAPPENS TO THE
BALL AFTER IT IS HIT?**

- It moves.
- It rolls in a certain direction.
- It stops only when something blocks it.

Descartes believed that the universe works like a giant machine where everything moves because objects push or collide with each other.

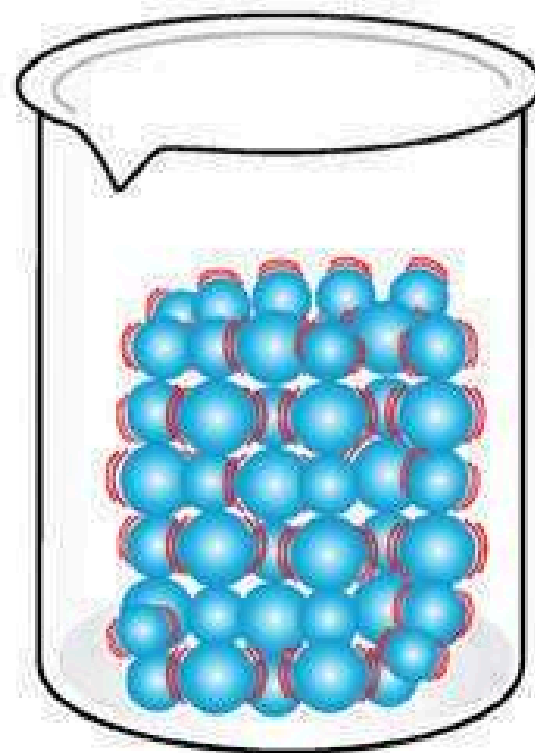




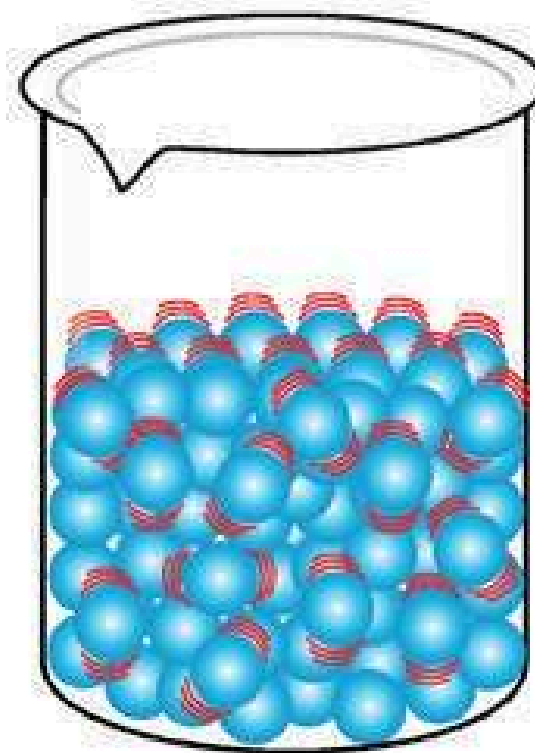
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Cartesian Physics

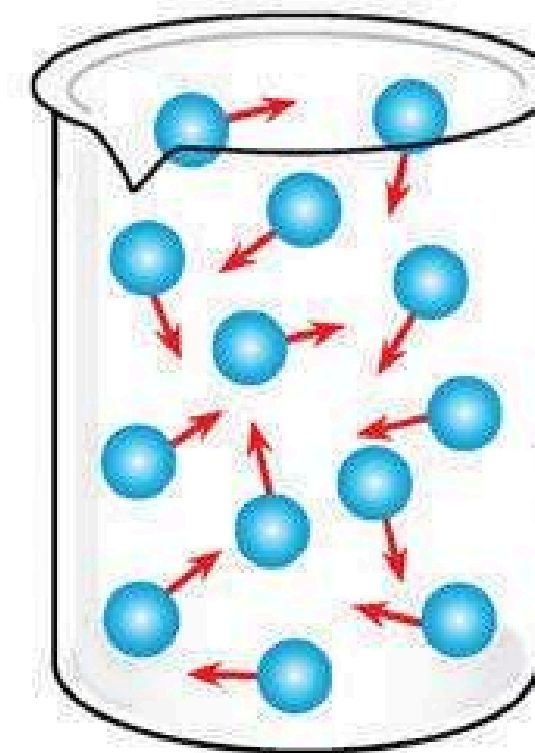
CARTESIANS BELIEVED THAT MATTER HAS ONE MAIN PROPERTY



Solid



Liquid



Gas



How do we know that something is matter?

- Before Descartes, philosophers had different answers to this question.

Ancient thinkers believed matter had multiple essential properties, such as:

- Shape
- Weight
- Color
- Hardness
- Temperature
- Purpose or natural tendency





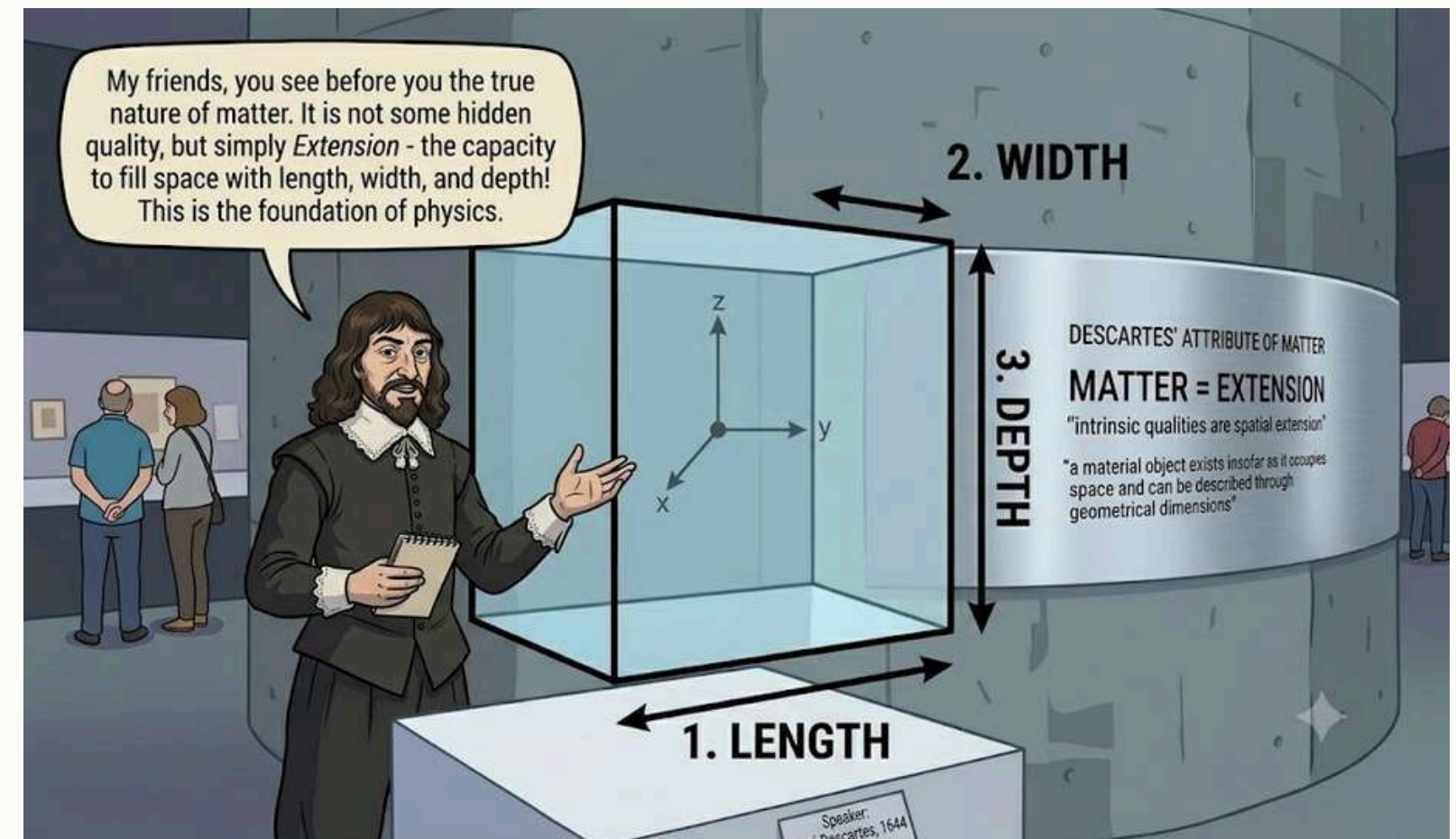
How do we know that something is matter?

René Descartes made an important distinction.

He argued that many properties we notice –such as:

- color
- smell
- taste
- texture

are not the true essence of matter because they depend partly on human perception.

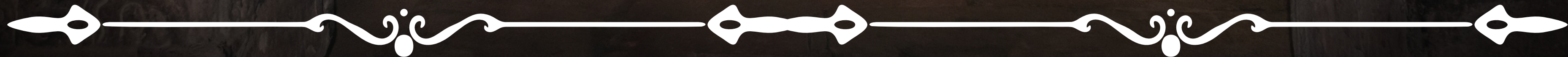


Descartes focused on what is truly intrinsic to matter: Extension



What is the essential nature of matter according to Descartes?

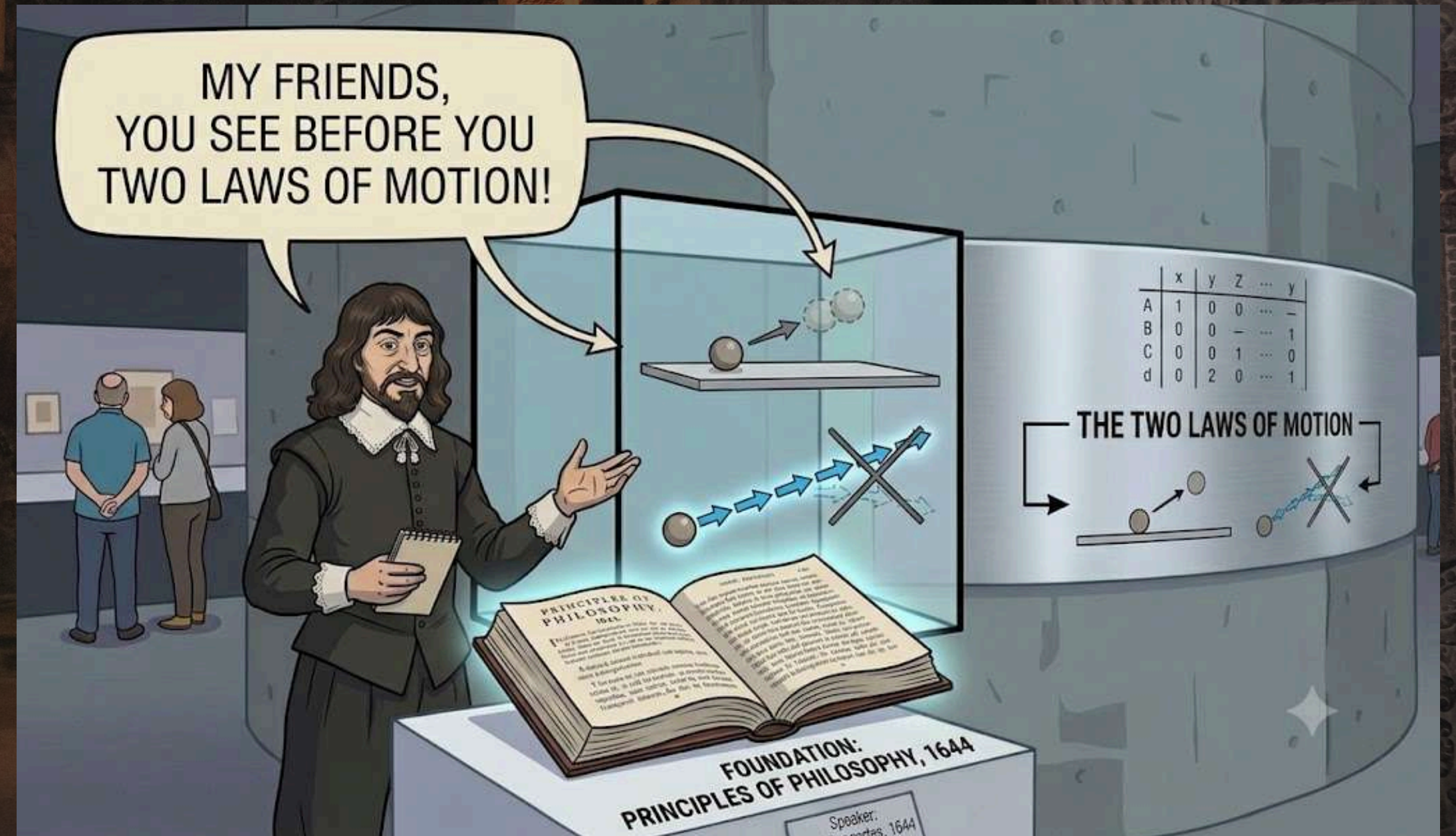
- **Extension:** The only property of material objects is that they occupy space (they have length, width, and depth). There are no hidden "magical" forces.
- **Action by Contact:** Because matter is just extended space, objects can only interact by physically touching or colliding with one another.





Descartes' Laws of Motion

From these foundational ideas, Descartes deduced his first two laws of motion in his 1644 text, *Principles of Philosophy*.





Descartes' Laws of Motion

The First Law of Motion (The Principle of Rest/Motion)

Every bit of matter maintains its state of rest or motion unless it collides with some other bit of matter.

- *The Metaphor: Think of billiard balls. A billiard ball sits perfectly still on a table until another ball strikes it. Once moving, it would run forever in a straight line if it didn't hit another ball or the edge of the table.*



IF AN OBJECT IS AT REST, IT WILL STAY AT REST,
UNLESS A NET FORCE ACTS ON IT.



Descartes' Laws of Motion

The Second Law of Motion (The Principle of Rectilinear Motion)

Every bit of matter, regarded by itself, tends to continue moving only along straight lines.

- *If unobstructed, an object's motion is always perfectly straight (rectilinear).*

IF AN OBJECT IS IN MOTION, IT WILL STAY IN MOTION, UNLESS A NET FORCE ACTS ON IT.





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Plenism & Vortex Theory

*Cartesians believed: Empty space does not exist.
Even outer space is filled with invisible matter.*

Is an “empty” bottle/glass really empty?

Why Motion Becomes Circular?





Plenism

If objects naturally want to move in straight lines, why do we constantly see circular or curved motion in nature (like the Moon orbiting the Earth)?

To solve this, Descartes introduced the concept of Plenism.

Derived from the Latin plenum ("full"). plenism states that the universe is completely full and contains no empty space.

- **Space is Matter:** For Descartes, "empty space" is an impossibility. Space is not a separate thing; it is simply a property (attribute) of matter.
- **The Invisible World:** A bedroom with no furniture isn't empty; it is entirely packed with invisible bits of matter (air particles).



Vortex Theory : Why Motion is Circular



- Because the universe is a tightly packed plenum, true straight-line motion is impossible in practice.
- If one particle moves, it cannot move into a void. It must physically displace the particle next to it, which displaces the next, creating a chain reaction.

Therefore, all actual motion in the universe consists of particles exchanging places in a circular dance. This is the foundation of Descartes' Vortex Theory.



Vortex Theory

Descartes used this mechanical whirlpool theory to explain how the entire solar system is organized without relying on an invisible pulling force like gravity:

The Sun/Stars:

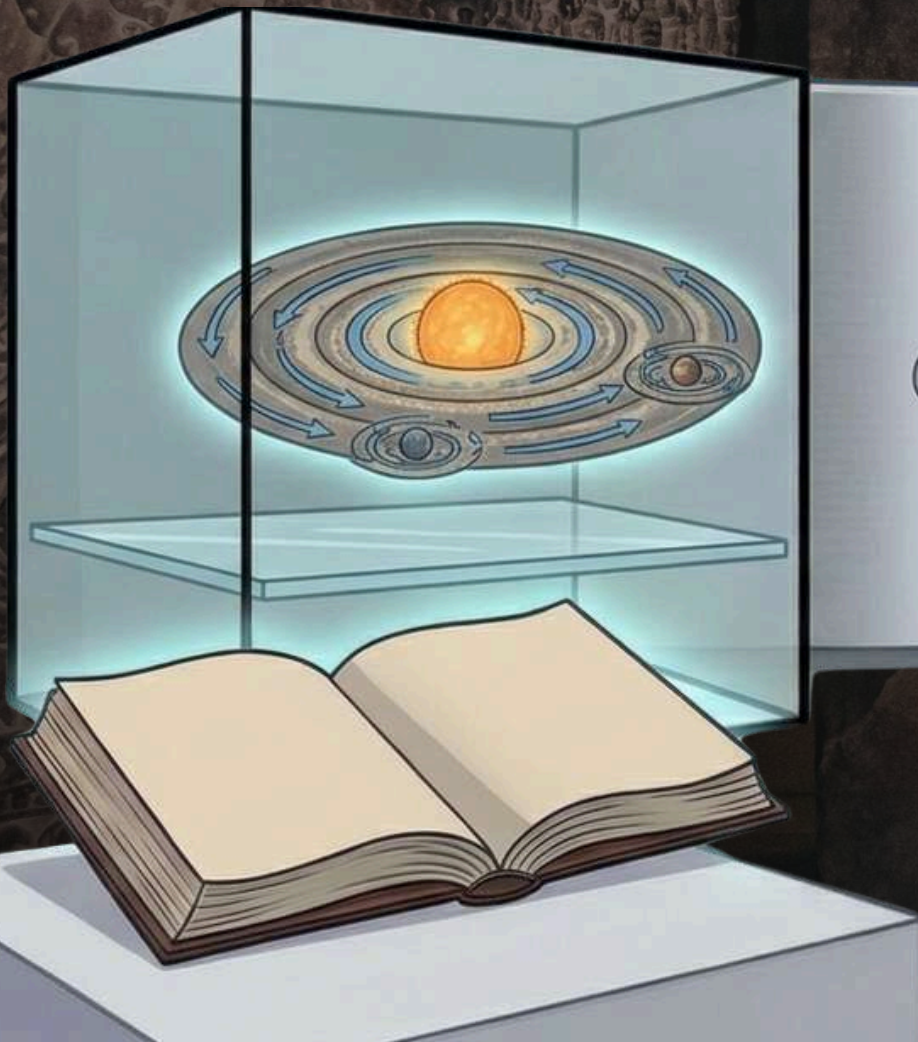
- Every star rotates on its axis, acting like a giant blender that stirs up the surrounding invisible matter into a massive whirlpool (vortex).

The Planets:

- Planets are caught up and carried around by this large solar vortex, much like leaves floating in a river whirlpool.

The Moons:

- As a planet rotates, it creates its own smaller, localized vortex within the larger one. This smaller whirlpool is what carries satellites (like our Moon) around the planet.





Mechanistic Explanations of Nature

A. Gravity as Inward Pressure

Cartesians did not believe in a mysterious pulling force called gravity.

Centrifugal force & Displacement

1. The Earth spins, creating a vortex of invisible matter extending past the moon.
2. The spinning action generates a centrifugal force, causing the particles to try to flee away from the center (the Earth).
3. Because the universe is a plenum (no empty space), these fleeing particles pack tightly at the outer edge of the vortex, creating a massive inward pressure.
4. The lightest, fastest particles make it to the outside, while this inward pressure forces the heavier, bulkier matter toward the center.
5. What we call "gravity" is actually this mechanical inward pressure pushing heavy objects down.



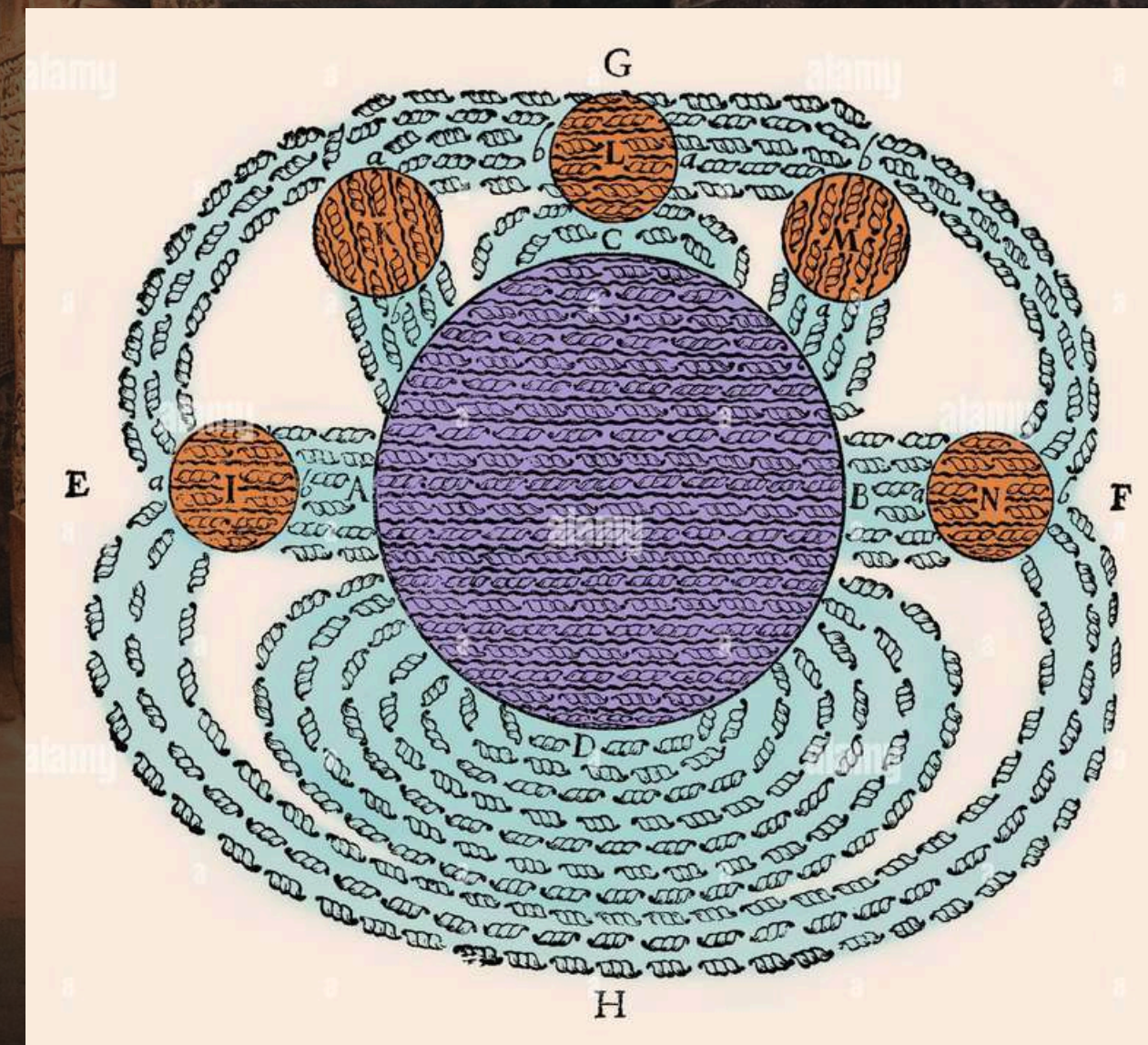


Mechanistic Explanations of Nature

B. Magnetism as Mechanical Corkscrews

Magnets appear to attract or repel iron from a distance without touching it. Descartes explained this using invisible plumbing:

- Every magnet (including the Earth) contains tiny, invisible, parallel tunnels called pores running from the North to the South pole. These pores are shaped like corkscrews.
- The Mechanism: A constant stream of invisible, corkscrew-shaped particles flows through these pores, loops around the outside of the magnet, and re-enters the other side.
- Attraction & Repulsion





Cartesian Method

Imagine this situation:

**You are alone at home. Suddenly, the door slams shut by itself.
What would be your first reaction?**

- Was it the wind?
- Did someone push it?
- Was it something supernatural?



Cartesian Method: Logical Explanations

Guide Question 1:

When something unusual happens, do you usually look for a logical explanation, or do you simply accept what you see?

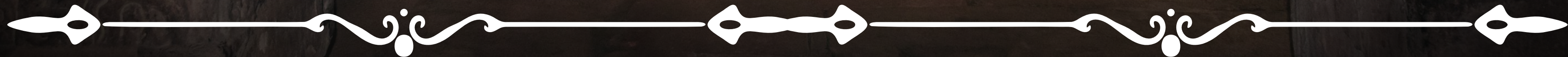
Searching for Causes

Most of us try to search for a cause. We ask: "What made that happen?"

This same question was important to philosophers and scientists like René Descartes.

The Cartesian Worldview

- Descartes believed that the universe works like a machine everything happens because of mechanical causes. This way of thinking became known as the Cartesian worldview.



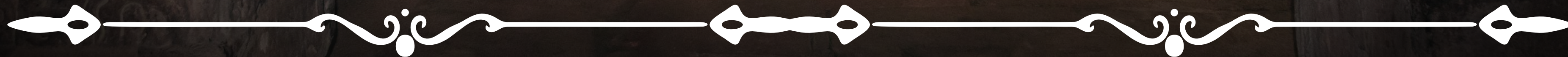


Aristotelian vs. Cartesian Thinking

Concept	Aristotelian View	Cartesian View
Explanation	Based on a thing's "nature."	Based on mechanical laws.
Example: Rock	Falls because heaviness is its natural property.	Falls because particles interact according to physical laws.

Which explanation sounds more scientific to you?

- A. It falls because that is its nature.
- B. It falls because forces and particles interact.

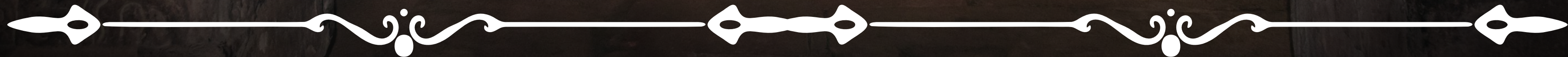




Cartesian Method

Because everything material occupies space, Descartes believed all physical things can be explained through:

- **Size:** The physical dimensions of the object.
- **Shape:** The geometric form of the matter.
- **Motion:** How the bits of matter move.
- **Arrangement of parts:** How components interact





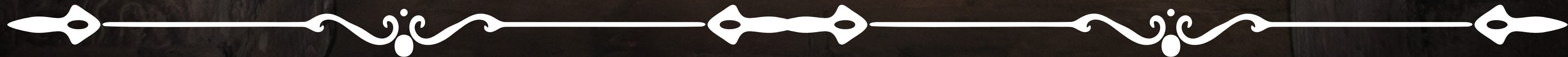
The Hypothetico-Deductive Method

René Descartes introduced a new way of studying nature called the Hypothetico-Deductive (HD) Method.

Instead of relying only on observation, scientists now needed to:

1. Make a hypothesis (an educated guess)
2. Use logic and reasoning to predict outcomes
3. Test the prediction through experiments
4. Accept the theory only if predictions are confirmed

This was a major change in science.

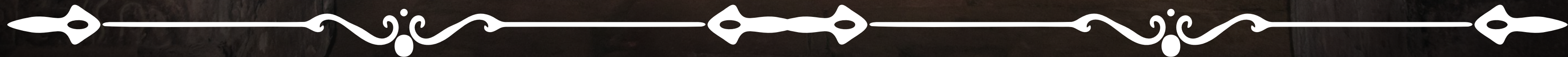




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The Hypothetico-Deductive Method





The Hypothetico-Deductive Method

Why Did This Change Happen?

Descartes believed that the main property of matter is extension.

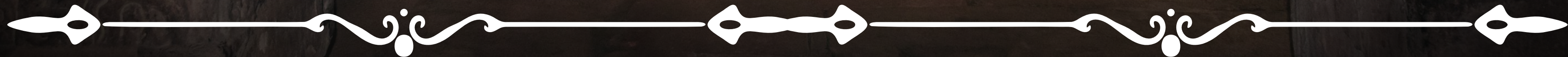
What is Extension?

- Extension means matter occupies space or takes up space.

Example:

- A book has size, width, and height, so it occupies space.

According to Descartes, everything physical is made of tiny particles that take up space.





The Hypothetico-Deductive Method

Hypothesis:

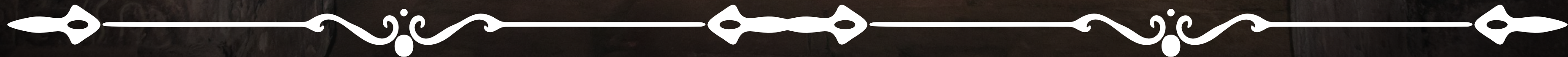
- Matter is extended substance (extension)

Deduction:

- If matter occupies space, then object cannot pass through each other.

Observation; Test:

- Indeed, two solid objects collide instead of merging.





1. Principle of Post-Hoc Explanations

- Because matter is just "bits of stuff taking up space," any single event can be explained by an infinite number of different invisible arrangements of those bits. (Since matter is made of tiny particles, one event can have many possible explanations.)

Example:

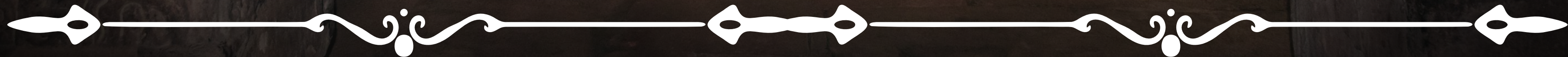
- Why did a glass break?

Possible explanations:

- It fell from the table
- Someone pushed it
- It had a crack already

A single observation does not always tell the full story.

We should not accept explanations too quickly.





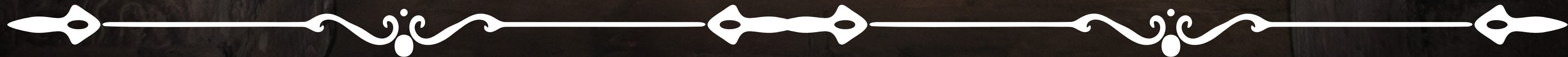
2. Principle of Complexity

The real world is far more complex beneath the surface than what our eyes actually see.

- The real world is more complex than what we see.
- What our eyes observe is only the surface.

Example:

- When water boils, we only see bubbles. But inside, tiny molecules are moving faster and colliding. This hidden activity explains what is happening.





The Three New Rules of Cartesian Science

1

Natural vs Artificial

Aristotelians separated natural and human-made things.
Cartesians rejected this. Everything follows the same laws.

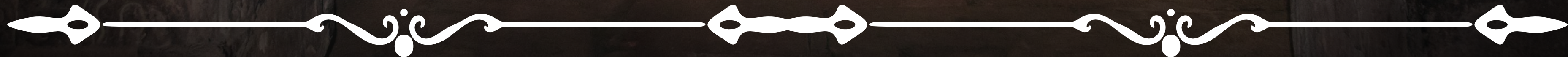
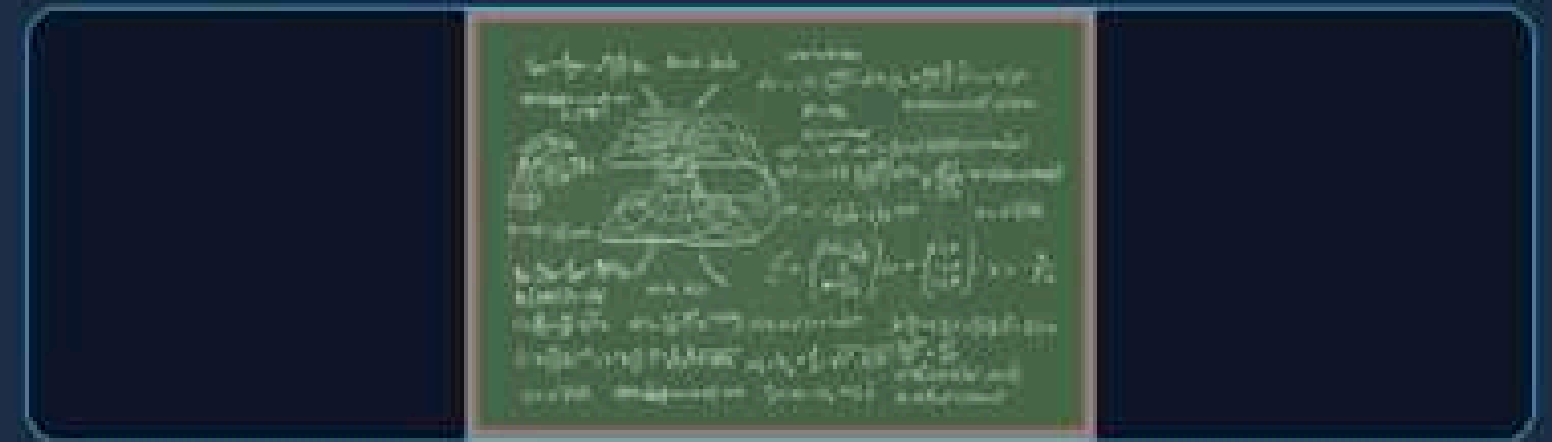
Guide Question 7: Why experiment? (Answer: Control variables and test accurately).

2

Mathematics is Universal

Can explain all material processes:

- Heart rate & population growth
- Genetics probabilities
- Enzyme reaction rates





The Three New Rules of Cartesian Science



"I think, therefore I am."

3

Mechanical Explanations Only

One major rule: All material events must have mechanical explanations using contact, movement, and interaction of parts.

Ex: A clock works because gears rotate and parts push other parts.

Guide Question 6: Why did they insist on this? (Answer: Logical, testable, observable).



Intuition

Where Intuition Was Allowed:

- Formal Sciences: Logic and mathematics.
- Example:
 - Descartes' famous "I think, therefore I am" (Cogito, ergo sum) to prove the mind exists.
 - $1 + 1 = 2$



Where Intuition Was Banned:

- Specific Physical Phenomena: You cannot use intuition to explain how gravity, magnetism, or biology works. Because you can imagine a dozen different mechanical ways a clock functions inside, you have to test your hypothesis against reality.



Conclusion

C - Change in worldview
A - Action by contact
R - Reason and rationality
T - Two substances (Dualism)
E - Experimentation and evidence
S - Science through mathematics

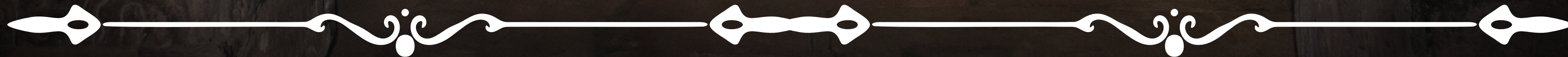
The CARTES worldview revolutionized science by promoting reason, experimentation, and mechanistic explanations, laying the foundation for modern scientific thought.



**Southern Luzon State University
Graduate School**



*What if everything you believe to be real is just
an illusion—how would you prove reality exists?*



Thank you

